

Solution
MAGNETISM -02
Class 10 - Science
Section A

1. (a) momentum

Explanation: We know when a proton moves in a magnetic field its velocity changes. Momentum is the product of mass and velocity, therefore momentum also changes. Velocity and momentum are the properties which change when a proton moves freely in a magnetic field.

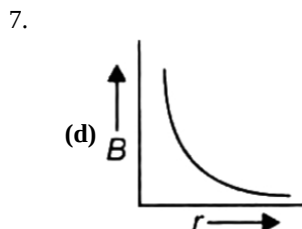
2. (b) Zero
Explanation: Magnetic field due to straight current carrying conductor is concentric circles which is parallel to the plane of the loop. Hence, force exerted is zero.

3. (b) $1.2 \times 10^{-3} \text{ Nm}$
Explanation: $1.2 \times 10^{-3} \text{ Nm}$

4. (a) $45\pi \times 10^{-7} \text{ T mA}^{-1}$
Explanation: $45\pi \times 10^{-7} \text{ T mA}^{-1}$

5. (a) An overload as well as short circuit.
Explanation: An electric fuse can disconnect the power supply from the circuit by causing a break in the circuit, thereby preventing the circuit from overload or short circuit. Thus heavy current is not allowed to flow through the circuit and the circuit and electric appliances are protected.

6. (d) Number of lines crossing a given point
Explanation: Number of lines crossing a given point



Explanation: The magnetic field strength (B) at the distance (r) from a long current carrying conductor is given by

$$B = \frac{\mu_0}{2\pi} \frac{I}{r}$$

For given current, $Br = \text{constant}$

i.e., as r increase, magnetic field B decrease.

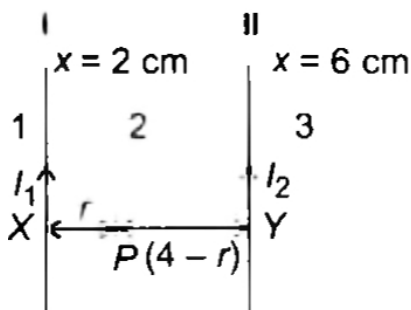
8. (a) Induced current will start flowing
Explanation: According to Lenz's law, when a closed coil is linked with a varying magnetic flux, then some current is induced in the coil.

So, when a magnet is taken towards a circular coil, then a varying magnetic flux is linked with the coil so that current is induced in the coil. This phenomenon is said to be electromagnetic induction.

9. (b) $56.4 \times 10^{-2} \text{ N}$
Explanation: $56.4 \times 10^{-2} \text{ N}$

10. (a) Electro magnetic Induction
Explanation: The transformer works on the principle of mutual induction of two coils or Faraday Law's Of Electromagnetic induction. When the current in the primary coil is changed the flux linked to the secondary coil also changes. Consequently, an EMF is induced in the secondary coil due to Faraday laws of electromagnetic induction.

11.
(d) Same at all points
Explanation: Same at all points
12.
(d) is same at all points.
Explanation: An even magnetic field exists inside a current carrying solenoid. The field lines are straight and parallel to each other.
13.
(d) $9.8 \times 10^{-3} \text{ T}$
Explanation: $9.8 \times 10^{-3} \text{ T}$
14.
(c) A, B and C
Explanation: **Magnets are made** from magnetic metals – iron, nickel and cobalt. These are the only pure metals that can be turned into a **permanent magnet**. **Steel** is an alloy of iron and so can also be **made** into a **magnet**. **Alnico** and Nipermag alloys are ferromagnetic, with a high coercivity (resistance to loss of magnetism) and are **used to make permanent magnets**.
15.
(b) Al, Ni, Co
Explanation: Alnico is the name for an iron alloy that primarily consists of iron (Fe), aluminum (Al), nickel (Ni), and cobalt (Co). That is why it is called al-ni-co. It can also be spelled AlNiCo. Sometimes, it may also contain Copper, Niobium, and Titanium. Alnico alloys are used as permanent magnets, as they are ferromagnetic.
16.
(a) Direction of current
Explanation: The direction of induced current in a straight conductor is given by Fleming's right-hand rule. The thumb points towards the direction of the current. Then fingers will wrap around the conductor in the direction of the field lines of the magnetic field.
17.
(b) Fleming's right hand rule
Explanation: The direction of induced current in a straight conductor is given by Fleming's right-hand rule. It states that if we stretch the thumb, forefinger and the middle finger of the right hand at right angles to one another in such a way that the forefinger points in the direction of the magnetic field. Then, thumb gives the direction of motion of conductor (force), forefinger indicates the direction of magnetic field, and the middle finger points the direction of induced current.
18.
(d) same at all points
Explanation: Inside the solenoid magnetic field lines are straight. This indicates a strong magnetic field. Hence the magnetic field is uniform at all points inside the solenoid.
19.
(b) 0.39 T
Explanation: 0.39 T
20.
(d) increases heavily
Explanation: A **short circuit** is simply a low resistance connection between the two conductors supplying electrical power to any **circuit**. This results in excessive current flow in the power source through the '**short**,' and may even cause the power source to be destroyed.
21.
(b) 3 cm mark
Explanation: Current carrying wires are shown in the figure. In region 1 and 3 magnetic fields produced by both the conductors have same direction so they will add on.



In region 2, magnetic fields produced by both the conductors are not same, so there is possibility to get a point where net magnetic field is zero.

Let P be the point where net magnetic field will be zero.

Here, $XP = r$ and $YP = (4 - r)$

$I_1 = 1 \text{ A}$, $I_2 = 3 \text{ A}$

Magnetic field produced by a current carrying conductor at a distance r from it,

$$B = \frac{\mu_0 I}{2\pi r}$$

$$\text{For conductor I, } B_I = \frac{\mu_0 I_1}{2\pi r}$$

$$\text{For conductor II, } B_{II} = \frac{\mu_0 I_2}{2\pi(4-r)}$$

Net magnetic field at P is zero. So, $B_I = B_{II}$

$$\frac{\mu_0 I_1}{2\pi r} = \frac{\mu_0 I_2}{2\pi(4-r)}$$

$$\Rightarrow (4 - r) I_1 = I_2 r$$

$$\Rightarrow 1(4 - r) = 3 r$$

$$\Rightarrow 4r = 4 \therefore r = 1 \text{ cm}$$

Position of the point P on the scale = $2 + 1 = 3 \text{ cm}$

22. (a) Solenoid

Explanation: A coil of many circular turn wrapped around a insulator forms a solenoid. It is mainly made of soft iron core as it loses the magnetism very easily on withdrawal of current.

23.

(b) Deflects towards the west.

Explanation: If we place a straight current-carrying wire (the direction of the flowing current is from north to south) such that the wire is placed below a compass needle such that the wire is parallel to the compass needle, the north pole of the compass needle deflects towards the west.

24. (a) Parallel straight lines

Explanation: Parallel straight lines

25.

(c) Current, length and distance of the wire

Explanation:

- The strength of the magnetic field depends upon the amount of current flowing in the wire, more current will produce a stronger field.
- Length of wire, the longer wire will produce stronger field.
- The distance of a point from the wire, more the distance of a point from the wire weaker the field.

26.

(c) Michael faraday

Explanation: Magnetic field lines were introduced by Michael Faraday (1791-1867) who named them "lines of force."

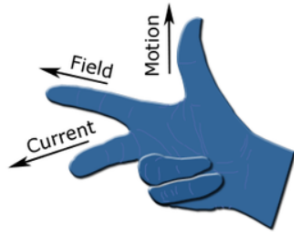
27.

(b) direction of the induced current

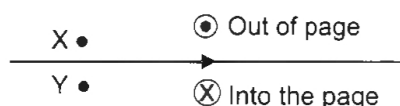
Explanation:

Fleming's right-hand rule shows the direction of the induced current when a conductor moves in a magnetic field. It can be used to determine the direction of current in a generator's winding. The thumb is pointed in the direction of motion of the

conductor.



28. (b) 15 A
Explanation: The power circuit with a 15 A fuse is used for running the electric heater, electric iron, geyser, refrigerator, etc. as it draws more current.
29. (b) red, blue, green
Explanation: The mains supply is delivered to houses using three core wiring called the live, neutral, and the earth. The live wire is red in colour and brings in the current, the neutral coloured blue is the return wire. The earth is green or yellow. These wire supply electricity to separate circuits within the house.
30. (a) direction of magnetic field
Explanation: Tangent indicates the direction of the magnetic field. It just an alternative used when compass is not available.
31. (a) out of the page
Explanation: The direction of force experienced by positron will be out of the page
32. (b) 36.36 A
Explanation: 36.36 A
33. (a) Option (c) and (d)
Explanation: Each moving charged particle in a magnetic field experiences a force. The direction of force experienced by a positive charge (i.e., a proton) is given by Fleming's left hand rule. The force acting on the proton would change both velocity and momentum.
34. (a) A.C. generator
Explanation: A.C. generator
35. (b) $H_1 < H_2$
Explanation: $H_1 < H_2$
36. (d) Oersted
Explanation: Hans Christian Oersted was a Danish Physicist who discovered that electric current can create magnetic fields which were the first connection between electricity and magnetism.
37. (d) Only A
Explanation: The magnitude of the magnetic field produced at a given point is directly proportional to the magnitude of the current passing through the conductor. The strength of the field at any point is inversely proportional to the distance of the point from the wire. ($B \propto I$ and $B \propto \frac{1}{r}$).
38. (d) The needles will deflect in the opposite directions.
Explanation: Current carrying wire produces magnetic field, but direction of magnetic field on the given points X and Y are different.



This is found by right hand thumb rule. So magnetic compasses on X and Y experience magnetic force of same magnitude but in opposite directions. Hence, the needles will deflect in opposite directions.

39.

(b) $\frac{\mu_0 NI}{2r}$

Explanation: The expression for the magnetic field due to the current at the centre of the coil is given by, $B = \frac{\mu_0 NI}{2r}$ where N is the number of turns all other has the usual meaning.

This magnetic field is perpendicular to the plane of the coil. If the current is clockwise the magnetic field is pointed inward to the plane, if the current is counter-clockwise it is pointed out of the plane.

40. (a) higher power ratings such as geyser.

Explanation: In domestic electric circuits the wiring with 15 A current rating is for the electric devices which have higher power ratings such as geyser.

41.

(d) 220 v, 50 Hz

Explanation: The voltage in India is 220 volts, alternating at 50 cycles (Hertz) per second. The advantage of 220 is lower current and thus less power loss. And the ability to use thinner wires - less copper.

42.

(c) Electromagnetic induction

Explanation: Electromagnetic induction

43. (a) fuse

Explanation: The most important safety, method used for protecting home appliances from short circuiting or overloading is the electric fuse. This is a safety device having thin wire of short length made of tin (25%) and lead (75%) alloy having low melting point around 200°C. The fuse wire is of chosen thickness, so as to fix its resistance and hence amount of heating on passage of a particular amount of current. Whenever current through the fuse exceeds the set limit, the fuse wire melts and breaks the circuit. This saves the main circuit components from damage.

44.

(b) $422.4 \times 10^{-7} \text{ T}$

Explanation: $422.4 \times 10^{-7} \text{ T}$

45. (a) V and Y

Explanation: V and Y

46.

(c) Fleming's left hand rule.

Explanation: Fleming's left hand rule.

47.

(b) Both the direction and the relative strength of magnetic field.

Explanation: Magnetic Field is the region around a magnet where other magnetic material will experience a force. A magnetic field can be graphically represented by magnetic field lines which indicates its strength and direction.

48. (a) 4f

Explanation: 4f

49.

(c) towards west

Explanation: The encircled cross represents the direction of the v electron. Applying the right-hand rule for the cross product of vectors, we find that the direction of the vector ($\mathbf{V} \times \mathbf{B}$) is towards east since $\mathbf{F}_m = -e(\mathbf{V} \times \mathbf{B})$, therefore the force is directed towards the west.

50.

(c) North pole

Explanation: According to clock rule if we look at the face of the coil which carries current in the clockwise direction then the face is called as south pole looking from the other side of coil we find that anti-clockwise current flows that is called as the north pole.

51.
(c) 5 ampere
Explanation: The lighting circuit with a 5 A fuse is used for running electric bulbs, fans, radio, TV, tube lights, etc.
52.
(a) upward
Explanation: The direction of the magnetic field can be determined by the Fleming's left hand rule. According to Fleming's left hand rule if we stretch the thumb, the centre finger and the middle finger of our left hand such that they are mutually perpendicular to each other. If the centre finger gives the direction of current and middle finger points in the direction of magnetic field then the thumb points towards the direction of the force or motion of the conductor. Here the direction of positively charged alpha particle is towards west this implies the direction of current is also towards west and the direction of magnetic force is towards north. Hence, according to Fleming's left hand rule, the direction of magnetic field will be upwards.
53.
(d) Current flowing through the solenoid is saturated.
Explanation: Current flowing through the solenoid is saturated.
54.
(c) Vertically downward on the wire
Explanation: By using Fleming's left hand rule, the force acts vertically downward.
55.
(c) Only (i) and (ii)
Explanation: When two parallel conductors conduct currents in opposite direction, the magnetic fields in between them produce forces that push them away.
56.
(c) Magnetic field
Explanation: Magnetic field
57.
(d) High resistance, Low melting point
Explanation: A fuse wire has high resistance and low melting point so that it will melt if a current of large magnitude passes through the wire.
58.
(c) Circular
Explanation: The nature of the magnetic field lines around a straight current-carrying conductor is concentric circles with centre at the axis of the conductor.
59.
(b) The field consists of concentric circles centred on the wire
Explanation: The field consists of concentric circles centred on the wire. It originates from the circumference of the wire and spread out thus creating concentric circles.
60.
(c) Is the same at all points
Explanation: The magnetic field inside a long straight current-carrying solenoid is uniform which is represented by parallel lines. Hence it is same at all the points.
61.
(b) Move out and into the page
Explanation: The direction of the current flowing in the conducting wire change alternatively when it is connected to an alternating current supply.
62.
(c) Alnico, Neodymium
Explanation: Materials used to make permanent magnets are Alnico an alloy of aluminium, nickel, cobalt. Neodymium magnets are widely used for computer hard disks. These magnets are the strongest.
63.
(b) An iron piece into it
Explanation: An iron piece into it

Section B

64. State True or False:

- (i) **(a)** True
Explanation: True
- (ii) **(a)** True
Explanation: True
- (iii) **(a)** True
Explanation: True
- (iv) **(a)** True
Explanation: True
- (v) **(a)** True
Explanation: True
- (vi) **(a)** True
Explanation: True
- (vii) **(a)** True
Explanation: True
- (viii) **(a)** True
Explanation: True
- (ix) **(b)** False
Explanation: False
- (x) **(a)** True
Explanation: True